Doob's Proof of MLE Strong Consistency: A Surprisal and Information Perspective

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Abstract

We give a concise, self-contained exposition of Doob's proof of strong maximum likelihood consistency, formulated in terms of surprisal and relative information. This formulation clarifies the roles of identifiability, properness, convergence, and continuity, and highlights that consistency holds without uniform convergence of the log-likelihood. We then apply the result to two special cases, i.i.d. sequences and stationary ergodic Markov processes, illustrated by an i.i.d. Gaussian sequence and an autoregressive Gaussian process.

Keywords: maximum likelihood estimator, true parameter, relative information, autoregressive, markov process, multivariate normal

1 Introduction

The consistency of the Fisher (1922) maximum likelihood estimator (MLE) is a classical result discovered more than a century ago. This method is typically treated as a special case of M-estimators, and derived via techniques dating back to Cramér (1946) and Wald (1949). A modern treatment is van der Vaart (1998). Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found the Doob (1934) strong consistency proof of MLE convergence more motivated and accessible to students, particularly for students already familiar with information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter spaces and general state spaces with essentially no extra effort only adds to its appeal.

The contribution of this note is expository and pedagogical: We give a formulation of Doob's consistency argument in surprisal/information terminology, in sufficient generality to encompass i.i.d. sequences and Markov processes as special cases. This is done without assuming uniform convergence. To highlight the applicability of our assumptions, we also work out two examples, l i.i.d. Gaussian sequences and autoregressive Gaussian processes.

2 Surprisal and Information

A random variable x is sampled repeatedly, resulting in samples[1] x_1, x_2, $\dots$. We assume the density $p(x,\theta^*)$ of x is one of a parametric family $p(x,\theta)$ of positive functions, and we wish to recover θ^* from x_1, x_2, $\dots$. While $p(x,\theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over the underlying expectation E_*. Given this, how do we recover θ^* from the process x_1, x_2, $\dots$?

Define the *surprisal* Samson (1951), Tribus (1961)

$$S(x,\theta) = \log(1/p(x,\theta)).$$

This terminology is intuitive: low probability equals high surprisal.

Assume $S(x,\theta)$ is integrable for all θ. Then it is natural to call a minimizer θ^* of the *expected surprisal* $E_*(S(x,\theta))$ a *true parameter*, and the corresponding density $p(x,\theta^*)$ a *true density*. Equivalently, given θ^* and θ, the *relative information* is

$$I(\theta^*,\theta) = E_*(S(x,\theta)) - E_*(S(x,\theta^*)). \tag{1}$$

The relative information is the excess expected surprisal of θ over θ^*. Then θ^* is a true parameter iff $I(\theta^*,\theta) \geq 0$ for all θ.

While $p(x,\theta)$ have so far been arbitrary positive functions, we now specialize to the case where $p(x,\theta)$ are densities in the usual sense. We say θ^* is a *reference parameter* if

$$E_*\left(\frac{p(x,\theta)}{p(x,\theta^*)}\right) = 1 \qquad \text{for all } \theta.$$

Then the corresponding density $p(x,\theta^*)$ is a *reference density*.

Let θ^* be a reference parameter, and define E and E_θ by

$$E_*(f(x)) = E(p(x,\theta^*)f(x)) \qquad \text{and} \qquad E_\theta(f(x)) = E(p(x,\theta)f(x)).$$

Then the relative information takes the familiar form Kullback & Leibler (1951), Sanov (1957), Donsker & Varadhan (1975), Dembo & Zeitouni (1998)

$$I(\theta^*,\theta) = E\left(p(x,\theta^*)\log\left(\frac{p(x,\theta^*)}{p(x,\theta)}\right)\right), \tag{2}$$

and $E_\theta(1) = 1$ for all θ. Since $j(p) = \log(1/p)$ is convex, by Jensen's inequality,

$$I(\theta^*,\theta) = E_*\left(j\left(\frac{p(x,\theta)}{p(x,\theta^*)}\right)\right) \geq j\left(E_*\left(\frac{p(x,\theta)}{p(x,\theta^*)}\right)\right) = j(E_\theta(1)) = j(1) = 0, \qquad \text{for all } \theta.$$

[1]We use the same letter (e.g. x) to denote both a random variable and a sample of it.

It follows that a reference density is a true density.

If θ is any other true parameter, then $I(\theta^*, \theta) = 0$, hence, by strict convexity of $j(p)$, there is a scalar t such that $p(x, \theta) = t \cdot p(x, \theta^*)$. Since θ^* is a reference parameter, $t = 1$, hence $p(x, \theta) = p(x, \theta^*)$: every true density equals the reference density. It follows that there is at most one reference density; when there is a reference density, it is the unique true density.

We say parameters θ, θ' *share a density* if $p(x, \theta) = p(x, \theta')$. If parameters do not share densities, there is at most one reference parameter; moreover, when there is a reference parameter, it is the unique true parameter.

The simplest example is as follows. A vector $\theta = (\theta_1, \theta_2, \dots, \theta_d)$ is a probability vector if its components are nonnegative and sum to one. Suppose $\theta^* = (\theta_1^*, \theta_2^*, \dots, \theta_d^*)$ is a probability vector and a random variable x satisfies $P_*(x = i) = \theta_i^*$, $i = 1, 2, \dots, d$. Then the mass function of x is $p(x, \theta^*) = \theta_x^*$. If θ is any other probability vector, the surprisal is $S(x, \theta) = \log(1/\theta_x)$, and

$$I(\theta^*, \theta) = \sum_{i=1}^{d} \theta_i^* \log(\theta_i^*/\theta_i). \tag{3}$$

In the form (2), $I(\theta^*, \theta)$ is more commonly known as Kullback-Leibler divergence van der Vaart (1998) or relative entropy Cover & Thomas (2006), Dembo & Zeitouni (1998) or both Bishop (2006), Murphy (2022). It appears as the rate function in Sanov's theorem Sanov (1957). It is natural to regard $I(\theta^*, \theta)$ as information, following some authors Kullback (1959), Schervish (1995), since information and entropy are conceptual opposites: the Shannon (1948) entropy $-\sum_{i=1}^{d} p_i \log p_i$ is concave, while (3) is convex.

Returning to the general setting, the *likelihood* and (empirical) *mean surprisal* are

$$L_n(\theta) = \prod_{k=1}^{n} p(x_k, \theta), \qquad S_n(\theta) = \frac{1}{n} \log(1/L_n(\theta)) = \frac{1}{n} \sum_{k=1}^{n} S(x_k, \theta).$$

The mean surprisal is also called the negative log-likelihood.

By definition, θ^* minimizes the expected surprisal. The MLE method rests on the convergence of the mean surprisal to the expected surprisal for large samples. While this is not the case in general, this convergence is guaranteed in two classical cases: an i.i.d. sequence Fisher (1922), and a stationary ergodic Markov process Anderson & Goodman (1957), Billingsley (1961). In the former case, the convergence is a consequence of the law of large numbers; in the latter case, of the ergodic theorem.

Because of this, it is natural to estimate θ^* by a minimizer of $S_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the samples, maximizing $L_n(\theta)$ over all θ.

In what follows, only E_*, $p(x, \theta)$, and true parameters enter; E, E_θ, and reference parameters play no role.

3 Consistency

In addition to integrability of $S(x, \theta)$, we make four assumptions. The first is

A (Identifiability)**.** *There is at most one true parameter.*

A follows if the expected surprisal is strictly convex. Also, as we saw above, A follows if there is a reference parameter and parameters do not share densities.

A sequence of parameters *subconverges* to θ if a subsequence converges to θ. If a sequence subconverges to θ, θ is a *limit point of the sequence.* A sequence has exactly one limit point θ iff the sequence converges to θ. A subset K of the parameter space is *compact* if every sequence in K has at least one limit point in K.

A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \leq c$ are compact subsets of the parameter space, for every level c. Since a proper continuous function has a minimizer and $\hat{\theta}_n$ should minimize $S_n(\theta)$, this leads to the second

B (Properness)**.** *There is an* $N \geq 1$*, depending on the samples, and a proper* $F(\theta)$*, not depending on the samples, such that, with probability one,*

$$F(\theta) \leq S_n(\theta), \qquad \text{for all } \theta \text{ and } n \geq N. \tag{4}$$

When the parameter space is compact, the proof below shows assumption B is superfluous. The assumption crucial to the method is the third

C (Convergence)**.** *For every integrable* $g(x)$

$$\frac{1}{n}\sum_{k=1}^{n} g(x_k) \to E_*(g(x)), \qquad \text{as } n \to \infty, \qquad \text{with probability one.} \tag{5}$$

We will need the convergence (5) with $g(x) = S(x,\theta)$ simultaneously for all θ, in the sense

$$S_n(\theta) \to E_*(S(x,\theta)), \qquad \text{as } n \to \infty, \qquad \text{for all } \theta\text{, with probability one.} \tag{6}$$

We emphasize this is not uniform convergence, since the rate of convergence may vary across parameters θ. For this we need the fourth

D (Continuity)**.** *The parameter space is separable and there is a continuous* $\delta(\theta,\theta')$ *and an integrable nonnegative* $g(x)$ *satisfying*

$$|S(x,\theta) - S(x,\theta')| \leq \delta(\theta,\theta')g(x) \qquad \text{and} \qquad \delta(\theta,\theta) = 0. \tag{7}$$

Under C, D, a standard separability argument establishes (6). Under D, the expected surprisal $E_*(S(x,\theta))$ is continuous in θ. If moreover B holds, then by sending $n \to \infty$ in (4), $F(\theta) \leq E_*(S(x,\theta))$ for all θ, hence the expected surprisal is proper. Hence there is a unique true parameter under A, B, C, and D. Similarly, under B, D, the mean surprisal is continuous in θ and proper, hence an MLE $\hat{\theta}_n$ exists for $n \geq N$, with probability one.

These assumptions are particularly tractable for *exponential families* Cover & Thomas (2006), Murphy (2022), Schervish (1995), van der Vaart (1998), in which $S(x,\theta)$ is essentially a sum of products of the form $f(\theta)g(x)$. In §4, we verify these assumptions for a multivariate normal i.i.d process and an autoregressive Gaussian process.

In (6), there is no presumption of uniform convergence. This is a key feature of Doob's proof. The argument instead combines properness, convergence (6), and the continuity estimate in (7) to control limit points of the MLE sequence.

Theorem (Consistency)**.** *Let* x_1*,* x_2*, ... be a process, and let* $p(x,\theta)$ *be a parametric family of positive functions. Under the above assumptions,*

1. *there is a unique true parameter* θ^**,*
2. *with probability one, an MLE sequence* $\hat{\theta}_n$ *exists for sufficiently large* n*,*
3. *with probability one, any MLE sequence converges to the true parameter,* $\hat{\theta}_n \to \theta^*$*.*

Proof. The first two items were derived above. For the third item, with $g(X)$ as in (7), fix samples X_1, X_2, ... for which (4), (5), and (6) hold. We show $\hat{\theta}_n \to \theta^*$ for these samples. By (4),

$$F(\hat{\theta}_n) \leq S_n(\hat{\theta}_n) \leq S_n(\theta^*), \qquad \text{for } n \geq N.$$

By (6), the right side of this last display is bounded as $n \to \infty$. Since $F(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space, and thus has limit points.

Suppose θ is a limit point of $\hat{\theta}_n$. Then $\hat{\theta}_n$ subconverges to θ. By (7),

$$S_n(\theta) - S_n(\theta^*) \leq S_n(\theta) - S_n(\hat{\theta}_n) \leq \delta(\theta,\hat{\theta}_n) \cdot \frac{1}{n}\sum_{k=1}^{n} g(x_k). \tag{8}$$

Since the left side of (5) is bounded as $n \to \infty$, and $\delta(\theta,\hat{\theta}_n)$ subconverges to zero, the right side of (8) subconverges to zero. By (1) and (6), the left side of (8) converges to $I(\theta^*,\theta)$, hence $I(\theta^*,\theta) \leq 0$. But $I(\theta^*,\theta) \geq 0$ since θ^* is a true parameter, hence $I(\theta^*,\theta) = 0$. Since there is only one true parameter, $\theta = \theta^*$. Thus θ^* is the only limit point, hence $\hat{\theta}_n \to \theta^*$ as $n \to \infty$. ∎

4 Examples

The first example is MLE for an i.i.d. Gaussian sequence, and the second example is MLE for an autoregressive Gaussian process. In the first example, the sequence is denoted z_1, z_2, $\dots$—anticipating its role as the innovations driving the autoregressive process of the second example—reserving x_1, x_2, $\dots$ for that Markov process.

4.1 I.I.D. Gaussian Sequence

Let μ be in $\mathbf{R}^d$, Q a positive definite[2] $d \times d$ matrix, and let $v \cdot w$ be the dot product in $\mathbf{R}^d$. With z in $\mathbf{R}^d$, the *normal density with parameter* $\theta = (\mu, Q)$ is $p(z, \theta) = e^{-S(z,\theta)}$ where

$$S(z, \theta) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2}(z - \mu) \cdot Q^{-1}(z - \mu).$$

This defines the parametric family. A random variable z in $\mathbf{R}^d$ is *normally distributed with parameter* $\theta^* = (\mu^*, Q^*)$ if its moment-generating function is

$$E_*\left(e^{v \cdot z}\right) = \exp\left(v \cdot \mu^* + \frac{1}{2} v \cdot Q^* v\right). \tag{9}$$

Let z_1, z_2, $\dots$ be independent and identically distributed copies of z. We show $S(z, \theta) = \log(1/p(z, \theta))$ is integrable and θ^* is the true parameter, and we verify A, B, C, and D. Assumption C is verified, as it is the law of large numbers (LLN).

Let $\theta' = (\mu', Q')$, $g(z) = 1 + |z|^2$ and let $\delta(\theta, \theta')$ equal half of

$$|\log \det Q - \log \det Q'| + \left|Q^{-1}\mu - (Q')^{-1}\mu'\right| + \left\|Q^{-1} - (Q')^{-1}\right\| + \left|\mu \cdot Q^{-1}\mu - \mu' \cdot (Q')^{-1}\mu'\right|.$$

Then $\delta(\theta, \theta')$ is continuous. By standard manipulations (7) follows, establishing D.

Proposition 1. *For $Q^* > 0$ and c scalar, let $\alpha = e^{-2c-1} \det(2\pi e Q^*)$ and*

$$F_*(Q, Q^*) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \operatorname{trace}(Q^{-1}Q^*), \qquad Q > 0.$$

Then $F_*(Q, Q^*)$ is minimized iff $Q = Q^*$, and $F_*(Q, Q^*) \le c$ implies $\alpha Q^* \le Q \le Q^*/\alpha$.

Proof. Let $h(\lambda) = \log \lambda + 1/\lambda$, $\lambda > 0$. Then $h(\lambda)$ is minimized iff $\lambda = 1$ and $h(\lambda)$ is proper in the sense $h(\lambda) \le \log(1/\alpha)$ implies $\alpha \le \lambda \le 1/\alpha$. Write $Q^* = S^2$ with $S > 0$, and define Z by $Q = SZS$. Suppose $F_*(Q, Q^*) \le c$ and let λ_1, λ_2, $\dots$, λ_d be the eigenvalues of Z. Then the proposition follows from

$$\log(1/\alpha) - 1 \ge 2F_*(Q, Q^*) - 2F_*(Q^*, Q^*) = \log \det Z + \operatorname{trace}(Z^{-1} - I) = \sum_{i=1}^{d} (h(\lambda_i) - 1). \qquad \square$$

Using the moment generating function,

$$E_*(S(z, \theta)) = F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu).$$

Since $S(z, \theta)$ is bounded below for each θ, $S(z, \theta)$ is integrable. By the proposition, (μ^*, Q^*) is the unique minimizer of the expected surprisal, establishing A.

Now let $0 < \epsilon < 1/2$ and

$$F_\epsilon(\theta) = F_*(Q, (1 - 2\epsilon)Q^*) + \frac{1}{2}(1 - \epsilon)(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu), \qquad \theta = (\mu, Q).$$

By the proposition with $(1 - 2\epsilon)Q^*$ replacing Q^*, if $F_\epsilon(\theta) \le c$, then Q is confined to a compact subset of positive definite matrices and, subsequently, μ is confined to a compact subset of $\mathbf{R}^d$. Hence $F_\epsilon(\theta)$ is proper.

[2] Appendix A.

With

$$Q_n(\mu) = \frac{1}{n}\sum_{k=1}^{n}(z_k - \mu)\otimes(z_k - \mu), \qquad \hat{\mu}_n = \frac{1}{n}\sum_{k=1}^{n} z_k, \qquad \hat{Q}_n = Q_n(\hat{\mu}_n),$$

let $\hat{\theta}_n = (\hat{\mu}_n, \hat{Q}_n)$. Then $\hat{\mu}_n$ and $\hat{Q}_n$ are the usual sample mean and the biased sample covariance.

By the LLN, with probability one, the matrix sequences

$$(\hat{\mu}_n - \mu^*)\otimes(\hat{\mu}_n - \mu^*) \to 0, \qquad Q^* - Q_n(\mu^*) \to 0, \qquad \text{as } n \to \infty.$$

If Q_n is any sequence of symmetric matrices with $Q_n \to 0$, then $\|Q_n\| \to 0$. Since $Q_n \le \|Q_n\|\, I$, given any $Q > 0$, there is $N \ge 1$ such that $Q_n \le Q$ for $n \ge N$. By choosing $Q = \epsilon^2 Q^*$ and $Q = \epsilon Q^*$, it follows there is $N^* \ge 1$, depending on the samples, such that, with probability one,

$$(\hat{\mu}_n - \mu^*)\otimes(\hat{\mu}_n - \mu^*) \le \epsilon^2 Q^*, \qquad Q_n(\mu^*) \ge (1-\epsilon)Q^*, \qquad \text{for } n \ge N^*. \tag{10}$$

Fix samples z_1, z_2, ... for which (10) holds. We establish (4) for these samples and $F(\theta) = F_\epsilon(\theta)$.

Let

$$f_0(\mu) = Q^* + (\mu - \mu^*)\otimes(\mu - \mu^*), \qquad f_\epsilon(\mu) = (1-\epsilon)f_0(\mu) - \epsilon Q^*.$$

Then $F_\epsilon(\theta) = F_*(Q, f_\epsilon(\mu))$. With $w = \hat{\mu}_n - \mu^*$ and $v = \mu - \mu^*$, using (15) in the appendix,

$$v\otimes w + w\otimes v \le \epsilon(\mu^* - \mu)\otimes(\mu^* - \mu) + \frac{1}{\epsilon}(\hat{\mu}_n - \mu^*)\otimes(\hat{\mu}_n - \mu^*) \le \epsilon f_0(\mu), \qquad \text{for all } \mu \text{ and } n \ge N^*.$$

Writing $z_k - \mu = (z_k - \mu^*) - v$, we have $Q_n(\mu) = Q_n(\mu^*) - (v\otimes w + w\otimes v) + v\otimes v$. Combined with the above inequalities, this leads to

$$Q_n(\mu) \ge f_\epsilon(\mu), \qquad \text{for all } \mu \text{ and } n \ge N^*.$$

In particular, since $f_\epsilon(\mu) \ge (1-2\epsilon)Q^*$, we have shown $\hat{Q}_n > 0$ for $n \ge N^*$.

Since $S_n(\theta) = F_*(Q, Q_n(\mu))$ and $Q_n(\mu) \ge \hat{Q}_n$ by least squares, by the proposition,

$$S_n(\theta) = F_*(Q, Q_n(\mu)) \ge F_*(Q, \hat{Q}_n) \ge F_*(\hat{Q}_n, \hat{Q}_n) = S_n(\hat{\theta}_n), \qquad \text{for all } \theta \text{ and } n \ge N^*.$$

Thus $\hat{\theta}_n$ is the MLE for $n \ge N^*$, and

$$S_n(\theta) = F_*(Q, Q_n(\mu)) \ge F_*(Q, f_\epsilon(\mu)) = F_\epsilon(\theta), \qquad \text{for all } \theta \text{ and } n \ge N^*.$$

This establishes (4) and completes the derivation of B.

4.2 Autoregressive Gaussian Process

Let A be a $d \times d$ matrix. With x_0, x_1 in $\mathbf{R}^d$, the *transition density with parameter* $\theta = (\mu, A, Q)$ is $p(x_0, x_1, \theta) = e^{-S(x_0,x_1,\theta)}$ where

$$S(x_0, x_1, \theta) = \frac{1}{2}\log\det(2\pi Q) + \frac{1}{2}(x_1 - Ax_0 - \mu)\cdot Q^{-1}(x_1 - Ax_0 - \mu).$$

This defines the parametric family.

Let $\theta^* = (\mu^*, A^*, Q^*)$, and assume A^* is stable.[3] Define

$$c^* = (I - A^*)^{-1}\mu^*, \qquad L^* = \sum_{n=0}^{\infty}(A^*)^n Q^*((A^*)^t)^n, \tag{11}$$

both well-defined by stability of A^*; as shown below, these are the values of the mean and covariance of x_0 needed to make x_0, x_1, x_2, ... (defined via the recurrence (12) below) stationary.

[3] Appendix A.

Let z be normally distributed with parameter (μ^*, Q^*), and let x_0 be normally distributed with parameter (c^*, L^*). The *innovations process* z_1, z_2, $\dots$ consists of i.i.d. copies of z, assumed independent of x_0. The *autoregressive Gaussian process (of order one), with parameter* $\theta^* = (\mu^*, A^*, Q^*)$, is the process x_0, x_1, x_2, $\dots$ generated by the recurrence

$$x_n = A^* x_{n-1} + z_n, \qquad n \geq 1. \tag{12}$$

Then z_n is independent of x_0, x_1, $\dots$, x_{n-1}, and x_0, x_1, x_2, $\dots$ is a Gaussian Markov process.

Let (x_0, x_1), (x_1, x_2), $\dots$, be the *pair process*. We show $S(x_0, x_1, \theta)$ is integrable and θ^* is the true parameter, and we verify A, B, C, and D for the pair process, assuming stability of A^*.

For the pair process, the likelihood and mean surprisal are

$$L_n(\theta) = \prod_{k=1}^n p(x_{k-1}, x_k, \theta), \qquad S_n(\theta) = \frac{1}{n}\log(1/L_n(\theta)) = \frac{1}{n}\sum_{k=1}^n S(x_{k-1}, x_k, \theta).$$

We verify C first. Because the series in (11) converges, we can go backward in time and present x_0 more explicitly. Let z_0, z_{-1}, z_{-2}, $\dots$ be a Gaussian i.i.d. process, with parameter (μ^*, Q^*), independent of z_1, z_2, $\dots$. Then $\dots, z_{-2}, z_{-1}, z_0, z_1, z_2, \dots$ is the *extended* innovations process. If we set

$$x_0 \equiv \sum_{k=0}^\infty (A^*)^k z_{-k},$$

then by (11) x_0 is normally distributed with parameter (c^*, L^*) and independent of z_1, z_2, $\dots$. Moreover iterating (12) and combining,

$$x_n = (A^*)^n x_0 + \sum_{k=1}^n (A^*)^{n-k} z_k = \sum_{k=-\infty}^n (A^*)^{n-k} z_k, \qquad n \geq 0. \tag{13}$$

Thus x_0, x_1, x_2, $\dots$ is the image of the extended innovations process under a shift-invariant map (see Appendix B for definitions). Since the extended innovations process is i.i.d., it is stationary and ergodic, hence x_0, x_1, x_2, $\dots$ is stationary and ergodic (Appendix B). By the ergodic theorem Billingsley (1961)

$$\frac{1}{n}\sum_{k=1}^n g(x_{k-1}, x_k) \to E_*(g(x_0, x_1)), \qquad \text{as } n \to \infty, \qquad \text{with probability one,}$$

for $g(x_0, x_1)$ integrable. This establishes C for the pair process.

For D, with $\theta' = (\mu', A', Q')$, extend the previous $g(z)$ to $g(x_0, x_1) = 1 + |x_0|^2 + |x_1|^2$, and extend the previous $\delta(\theta, \theta')$ by three additional terms

$$\left\|Q^{-1}A - (Q')^{-1}A'\right\| + |A^t Q^{-1}\mu - (A')^t (Q')^{-1}\mu'| + \left\|A^t Q^{-1} A - (A')^t (Q')^{-1}(A')\right\|.$$

This establishes D.

For A, the mean and variance of $x_1 - Ax_0 - \mu = z_1 + (A^* - A)x_0 - \mu$ are

$$c^* - Ac^* - \mu = (A^* - A)c^* - (\mu - \mu^*) \qquad \text{and} \qquad Q^* + (A^* - A)L^*(A^* - A)^t.$$

Here we used $c^* - A^*c^* = \mu^*$ (11). If we let

$$f_0(\mu, A) = Q^* + (A^* - A)L^*(A^* - A)^t + ((A^* - A)c^* - (\mu - \mu^*)) \otimes ((A^* - A)c^* - (\mu - \mu^*)),$$

we conclude

$$E_*(S(x_0, x_1, \theta)) = F_*(Q, f_0(\mu, A)) \equiv F_0(\mu, A, Q) \tag{14}$$

is the sum of three terms. Since the first term is $F_*(Q, Q^*)$, $Q = Q^*$ is its unique minimizer. Since $L^* > 0$, $A = A^*$ is the unique minimizer of the second term, hence $(\mu, A) = (\mu^*, A^*)$ is the unique minimizer of the sum of the second and third terms. This establishes A. Moreover, since the second and third terms are nonnegative, inspecting $F_0(\mu, A, Q) \leq c$ implies Q, then A, then μ are confined to compact subsets of their domains, hence $F_0(\mu, A, Q)$ is proper.

5 Concluding Remarks

6 Disclosures

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A Linear Algebra

This is a review of the linear algebra used above. Let Q and Z be symmetric matrices. We say Q is *positive definite*, and write $Q > 0$, if $v \cdot Qv > 0$ for all vectors $v \neq 0$. We say Q is *nonnegative definite*, and write $Q \geq 0$, if $v \cdot Qv \geq 0$ for all vectors v. We write $Q \geq Z$ if $Q - Z \geq 0$. Then $aI \leq Q \leq bI$ iff the eigenvalues λ of Q satisfy $a \leq \lambda \leq b$. In particular, if $Q \geq 0$, the eigenvalues satisfy $\lambda \geq 0$, and, if $Q > 0$, the eigenvalues satisfy $\lambda > 0$. From this, if $Q \geq 0$, there is $S \geq 0$ satisfying $Q = S^2$, and $S > 0$ if $Q > 0$.

Given vectors $v = (s_1, s_2, \dots, s_d)$ and $w = (t_1, t_2, \dots, t_d)$, the *tensor product* $v \otimes w$ is the matrix whose ij-th entry is $s_i t_j$. Then $u \otimes u \geq 0$. In particular, taking $u = \sqrt{\epsilon}v - w/\sqrt{\epsilon}$, $u \otimes u \geq 0$ implies

$$v \otimes w + w \otimes v \leq \epsilon v \otimes v + \frac{1}{\epsilon} w \otimes w. \tag{15}$$

If $u_1, u_2, \dots, u_d$ is any orthonormal basis and $e_1, e_2, \dots, e_d$ is the standard one-hot encoded basis, then $\sum_{i=1}^d e_i \otimes e_i$ and $\sum_{i=1}^d u_i \otimes u_i$ both equal I.

For any square matrix A, the *trace* of A is the sum $\operatorname{trace}(A) = \sum_{i=1}^d e_i \cdot Ae_i$ of the diagonal entries of A. Then $\operatorname{trace}(QZ) = \operatorname{trace}(ZQ)$ and $\operatorname{trace}(Q(u \otimes u)) = u \cdot Qu$ follow immediately. If $u_1, u_2, \dots, u_d$ is an orthonormal basis of eigenvectors of Q, $\operatorname{trace}(Q)$ is the sum $\sum_{i=1}^d u_i \cdot Qu_i$ of the eigenvalues of Q, hence $\operatorname{trace}(Q) \geq 0$ if $Q \geq 0$. Since $Z \geq 0$ and S symmetric imply $SZS \geq 0$, $\operatorname{trace}(QZ) = \operatorname{trace}(S^2 Z) = \operatorname{trace}(SZS) \geq 0$. It follows the trace is *monotone*,

$$Q \geq S \text{ and } Z \geq 0 \qquad \implies \qquad \operatorname{trace}(QZ) \geq \operatorname{trace}(SZ).$$

The *norm* squared $\|Q\|^2 = \operatorname{trace}(Q^2)$ is the sum of the squares of the eigenvalues of Q, hence $Q \leq \|Q\|\, I$. Since $\|Q\|^2$ is the sum of the diagonal entries of Q^2, $\|Q\|^2$ is the sum of the squares of the entries of Q, and we have $|Ax| \leq \|A\|\,|x|$ and the *Cauchy-Schwarz inequality* $|\operatorname{trace}(QZ)| \leq \|Q\|\,\|Z\|$.

The *determinant* of Q is the product $\det Q$ of the eigenvalues of Q. Then $Q > 0$ implies $\det Q > 0$ and the determinant is *multiplicative*,

$$Z > 0 \text{ and } S > 0 \qquad \implies \qquad \det(SZS) = \det S \cdot \det Z \cdot \det S.$$

For this restricted setting, multiplicativity is a consequence of the *matrix determinant lemma* (Hijab 2026, Exercise 3.2.18). By reducing to eigenvalues, for Z symmetric, one has

$$1 + |\det(I+Z) - 1| \leq \exp\left(\sqrt{d}\,\|Z\|\right).$$

This implies $\det Q$ is continuous at $Q = I$. Writing $Q = S^2$ with $S > 0$, multiplicativity extends this to continuity at all $Q > 0$.

A matrix A is *stable* if its (possibly complex) eigenvalues satisfy $|\lambda| < 1$. If $\rho(A) = \max|\lambda|$ is the radius of the smallest disk containing the eigenvalues, Gelfand's formula states $\rho(A) = \lim_{n\to\infty} \|A^n\|^{1/n}$. If $\rho(A) < r < 1$, then $\|A^n\| \leq r^n$ eventually, so the geometric series $\sum A^n$ converges, and hence $(I - A)^{-1}$ exists. Similarly, the Lyapunov series in (11) also converges.

B Ergodicity

For (5), we are interested in two cases, when the process is an i.i.d. sequence, and when it is a Markov process. In the former case, (5) is the strong law of large numbers (LLN). In the latter case, it is a consequence of the ergodic theorem.

The first step in deconstructing (5) is to take the expectation of its left side. For this to equal the right side, it is simplest to assume the marginals $E_*(g(x_n))$ are all equal: A process $x_1, x_2, \dots$ is stationary if expectations are unchanged under the shift $x_n \to x_{n+1}$. More precisely, the process is *stationary* if, for any function $G(x_1, x_2, \dots)$ of the process,

$$E_*(G(x_1, x_2, \dots)) = E_*(G(x_{n+1}, x_{n+2}, \dots)), \qquad \text{for every } n \geq 1. \tag{16}$$

Here the function G may be a function of a single sample $G(x_1)$, or two samples $G(x_1, x_2)$, or all the samples $G(x_1, x_2, \dots)$. For example, because of the limit, the left side of (5) is a function of all the samples. In practice the process generating the samples may not be stationary, and assuming stationarity is considered a simplifying first step.

Let $\bar{g}$ denote the left side of (5), when the limit exists. Since shifting the index by one $x_n \to x_{n+1}$ changes only the first and last terms of a partial average, these discrepancies vanish in the limit, and $G = \bar{g}$ satisfies (16) with E_* deleted:

$$G(x_1, x_2, \dots) = G(x_{n+1}, x_{n+2}, \dots), \qquad \text{for every } n \geq 1. \tag{17}$$

A function $G(x_1, x_2, \dots)$ satisfying this is called *invariant*. Thus, if the limit in (5) exists, it is an invariant function.

The ergodic theorem Birkhoff (1931), Durrett (2019) guarantees the limit in (5) exists for every stationary process. A process is *ergodic* if every invariant function is a constant, with probability one. The second step is to assume ergodicity. When this happens, the limit in (5) is a constant, hence, by the first step, equal to the expectation $E_*(g(x_1))$. We conclude assumption C holds when $x_1, x_2, \dots$ is stationary and ergodic.

We specialize ergodicity to two classes of processes, i.i.d. sequences and Markov processes. Because the Markov processes we consider are built out of i.i.d. sequences, we denote the i.i.d. sequences by $z_1, z_2, \dots$, reserving the notation $x_1, x_2, \dots$, for the Markov processes.

Suppose $z_1, z_2, \dots$ are independent and identically distributed. Then (16) follows immediately, yielding stationarity. For such a sequence, ergodicity is particularly simple to verify, and follows from the Kolmogorov zero-one law Kolmogorov (1933, 1950), Durrett (2019). Indeed, since an invariant function does not depend on $z_1, z_2, \dots, z_n$ for any n, it is a tail event, and Kolmogorov's zero-one law states every tail event has probability zero or one — hence every invariant function is constant with probability one, i.e., the process is ergodic.

The Markov processes considered are built from recurrences driven by an i.i.d. sequence. For such processes, to derive ergodicity, it is simplest to present them as shift-invariant functions of the i.i.d. sequence. In the form we need it, we'll need doubly-infinite processes.

A doubly-infinite process is a process $\dots, z_{-2}, z_{-1}, z_0, z_1, z_2, \dots$. Suppose x_0 is a function of a doubly-infinite process

$$x_0 = F(\dots, z_{-2}, z_{-1}, z_0, z_1, z_2, \dots).$$

Let σ be the shift, sending $x_n \to x_{n+1}$. Then we may define another doubly-infinite process x by

$$x_n = F(\sigma^n(\dots, z_{-2}, z_{-1}, z_0, z_1, z_2, \dots)) = F(\dots, z_{n-2}, z_{n-1}, z_n, z_{n+1}, z_{n+2}, \dots). \tag{18}$$

Now assume the z process is stationary. Then the law of the z process equals the law of the z process shifted. Since shifting z corresponds to shifting x in (18), the x process is stationary. Moreover, if $G(\dots, x_{-2}, x_{-1}, x_0, x_1, x_2, \dots)$ is an invariant function of x, then, from (18), it is an invariant function of z: substituting (18), G becomes a function of the z process alone; since shifting x corresponds to shifting z, invariance of G under the x-shift is exactly invariance under the z-shift. We have derived

Proposition 2. *If z is a stationary and ergodic process and (18) holds, then x is stationary and ergodic.*

Since an invariant function of the one-sided process $x_0, x_1, \dots$ is, in particular, an invariant function of the doubly-infinite process x (one that happens not to depend on $x_{-1}, x_{-2}, \dots$) proposition 2 also gives stationarity and ergodicity of $x_0, x_1, x_2, \dots$.

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